AP BIOLOGY — SCHOLARLY COMPENDIUM & REFERENCE SHEETS

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Comprehensive Academic Edition across all 8 CED Units & 60 Topics — Featuring Hierarchical Section Concept Maps, Formal Definitions Lexicons, Vector Diagrams, FRQ Scoring Templates, Experimental Design, Worked Quantitative Math, and Clinical/Ecological Case Studies.

8	60	300+
CED UNITS	MONOGRAPHS	CONCEPT MAPS

Search any AP Biology topic, mechanism, or equation (e.g., 'Photosynthesis', '1.5', 'Hardy-Weinberg', 'Operon')...

UNIT I | UNIT II | UNIT III | UNIT IV | UNIT V | UNIT VI | UNIT VII | UNIT VIII

UNIT I: CHEMISTRY OF LIFE

8–11% AP EXAM WEIGHT

1.1 Structure of Water and Hydrogen Bonding

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1.2 Elements of Life

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1.3 Introduction to Biological Macromolecules

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1.4 Properties of Biological Macromolecules (Carbohydrates)

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1.5 Structure & Function of Macromolecules (Lipids)

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1.6 Nucleic Acids

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1.7 Proteins

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UNIT II: CELL STRUCTURE AND FUNCTION

10–13% AP EXAM WEIGHT

2.1 Cell Structure: Subcellular Components

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2.2 Cell Structure and Function (Cell Size & SA:V)

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2.3 Plasma Membrane (Fluid Mosaic Model)

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2.4 Plasma Membrane Permeability

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2.5 Membrane Transport Overview

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2.6 Facilitated Diffusion (Channels, Carriers, Aquaporins)

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2.7 Tonicity and Osmoregulation (Water Potential Ψ)

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2.8 Mechanisms of Transport (Pumps & Cotransport)

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2.9 Cell Compartmentalization

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2.10 Origins of Cell Compartmentalization (Endosymbiosis)

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UNIT III: CELLULAR ENERGETICS

12–16% AP EXAM WEIGHT

3.1 Enzyme Structure and Catalysis

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3.2 Environmental Impacts on Enzyme Function & Inhibition

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3.3 Cellular Energy (Thermodynamics & ATP Coupling)

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3.4 Photosynthesis (Light Reactions & Calvin Cycle)

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3.5 Cellular Respiration & Fermentation

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UNIT IV: CELL COMMUNICATION AND CELL CYCLE

10–15% AP EXAM WEIGHT

4.1 Cell Communication (Signaling Modes & Quorum Sensing)

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4.2 Introduction to Signal Transduction (GPCR, RTK, Intracellular)

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4.3 Signal Transduction Pathways (cAMP & Kinases)

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4.4 Feedback Mechanisms (Negative vs. Positive Feedback)

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4.5 Cell Cycle and Mitosis

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4.6 **Regulation of Cell Cycle (Checkpoints, Cyclin-Cdk, p53)**

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UNIT V: HEREDITY

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5.1 **Meiosis (Reductional & Equational Division)**

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5.2 **Meiosis and Genetic Diversity (Crossing Over & Nondisjunction)**

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5.3 **Mendelian Genetics, Probability & Chi-Square (χ^2)**

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5.4 **Non-Mendelian Genetics, Linkage Maps & Maternal Inheritance**

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5.5 **Environmental Effects on Phenotype (Phenotypic Plasticity)**

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UNIT VI: GENE EXPRESSION AND REGULATION

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6.1 **DNA and RNA Structure**

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6.2 **DNA Replication (Replication Fork Enzymes)**

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6.3 **Transcription and RNA Processing (Cap, Tail, Splicing)**

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6.4 **Translation (Codons, tRNA, Ribosome A/P/E Sites)**

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6.5 Regulation of Gene Expression (lac/trp Operons & Epigenetics)

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6.6 Gene Expression and Cell Specialization (Transcription Factors)

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6.7 Mutations & Horizontal Gene Transfer

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6.8 Biotechnology (PCR, Gel Electrophoresis, Transformation, CRISPR)

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UNIT VII: NATURAL SELECTION

13–20% AP EXAM WEIGHT

7.1 Introduction to Natural Selection & Evolutionary Fitness

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7.2 Natural Selection Modes (Directional, Stabilizing, Disruptive)

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7.3 Artificial Selection & Selective Breeding

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7.4 Population Genetics (Genetic Drift, Bottleneck, Founder, Gene Flow)

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7.5 Hardy-Weinberg Equilibrium ($p + q = 1$, $p^2 + 2pq + q^2 = 1$)

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7.6 Evidence of Evolution (Homology, Fossils, Molecular Clocks)

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7.7 Common Ancestry (LUCA & Shared Eukaryotic Features)

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7.8 Continuing Evolution (Antibiotic Resistance & Viral Evolution)

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7.9 Phylogeny and Cladograms

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7.10 Speciation (Allopatric/Sympatric & Reproductive Barriers)

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7.11 Variations in Populations & Extinction

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7.12 Origins of Life on Earth (Miller-Urey & RNA World)

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UNIT VIII: ECOLOGY

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8.1 Responses to the Environment (Taxis, Kinesis, Tropisms, Signaling)

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8.2 Energy Flow Through Ecosystems (10% Rule, Endotherms/Ectotherms)

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8.3 Population Ecology (Exponential vs. Logistic Growth Equations)

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8.4 Effect of Density of Populations (Carrying Capacity K)

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8.5 Community Ecology (Symbiosis, Niches, Simpson's Index)

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8.6 Biodiversity, Keystone Species & Trophic Cascades

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8.7 **Disruptions to Ecosystems (Invasives, Eutrophication, Biomagnification)**

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